Floquet Theory

and Periodic Linear Systems

Lecture Notes — Complete Edition
With Fully Solved Exercises

Topics covered:

Periodic ODEs · Matrix Logarithms · Floquet Normal Form
Monodromy Matrices · Floquet Multipliers & Exponents
Hill's Equation · Lyapunov's Theorem · Stability Theory

Version 1.0 — Complete with Solved Exercises

Contents

Preface **iii**

1 Scalar Periodic ODEs **1**
1.1 The Scalar Equation and Periodic Solutions 1

2 The Matrix Exponential and the System $x' = Ax$ **3**
2.1 Constant-Coefficient Systems 3

3 Matrix Logarithms and Square Roots **4**
3.1 Square Roots of Real Matrices 4

4 The Complex Floquet Theorem **6**
4.1 Fundamental Matrices 6
4.2 The Complex Floquet Decomposition 6
4.3 The Real Floquet Theorem 7

5 The Floquet Transformation **8**
5.1 The Monodromy Matrix via the Floquet Decomposition 9

6 Monodromy Matrices and Floquet Multipliers **10**
6.1 Definitions 10
6.2 Characterisation of Monodromy Matrices 10
6.3 Conjugacy of Monodromy Matrices 11
6.4 Floquet Solutions 11

7 Liouville's Theorem and the Floquet Product Formula **13**

8 Stability and Boundedness **14**
8.1 Unbounded Solutions 14

9 Hill's Equation **16**
9.1 Reduction to a First-Order System 16

10 The Lyapunov Boundedness Theorem **17**
10.1 The Lyapunov–Borg Inequality 17
10.2 Lyapunov's Stability Theorem for Hill's Equation 18

11 Inhomogeneous Periodic Systems **20**
11.1 Existence and Uniqueness of Periodic Solutions 20
11.2 Bounded Solutions and Periodic Solutions 21

12 Solved Exercises **22**
Exercise 1: Periodicity of $x' = a(t)x$ 22
Exercise 2: Scalar periodicity condition 22
Exercise 3: Floquet decomposition for $n = 1$ 23
Exercise 4: T-periodic iff $X(0) = X(T)$ 23
Exercise 5: Why ruling out real Floquet multipliers implies boundedness . 23
Exercise 6: Linear independence for distinct multipliers 24
Exercise 7: Recovering the multiplier from a single component 24
Exercise 8: Monodromy for constant-coefficient system 25
Exercise 9: Product formula application 25

Index of Notation **26**

Preface

These notes develop the classical *Floquet theory* of linear periodic ordinary differential equations, beginning with the scalar case and progressing through the full n-dimensional matrix theory. All results are stated precisely with complete proofs, and every exercise that appears in the original notes is solved in full detail immediately after its statement.

The central objects of study are *fundamental matrices* of the system

$$x'(t) = A(t)\,x(t), \qquad A \in C(\mathbb{R};\, M_n(\mathbb{K})), \quad A \text{ is } T\text{-periodic},$$

together with their associated *Monodromy matrices*, *Floquet multipliers*, and *Floquet exponents*.

Notation. Throughout, $\mathbb{K}$ denotes either $\mathbb{R}$ or $\mathbb{C}$; $M_n(\mathbb{K})$ is the algebra of $n \times n$ matrices over $\mathbb{K}$; $GL_n(\mathbb{K})$ is the group of invertible such matrices; $\sigma(A)$ is the spectrum of A; and $C^k(I, V)$ is the space of k-times continuously differentiable functions from interval I to a Banach space V.

Chapter 1

Scalar Periodic ODEs

1.1 The Scalar Equation and Periodic Solutions

We begin with the simplest setting: a first-order linear ODE with a periodic, scalar coefficient.

Example 1.1 (Scalar periodic equation)**.** Let $a \in C(\mathbb{R})$ be T-periodic (i.e. $a(t+T) = a(t)$ for all t). Consider the equation

$$x'(t) = a(t)\,x(t). \tag{P$_1$}$$

The unique solution with initial condition $x(t_0) = x_0$ is

$$x(t) = x_0 \exp\left[\int_{t_0}^{t} a(s)\,ds\right].$$

Specific instances:

(a) $a(t) = 1$: then $x(t) = x(t_0)\,e^{t-t_0}$, which is *not* periodic (it grows without bound).

(b) $a(t) = \cos t$: then $x(t) = x(t_0)\,e^{\sin t - \sin t_0}$, which *is* periodic (since $\sin t$ is periodic with the same period as $\cos t$).

Theorem 1.2 (Characterisation of periodic solutions of the scalar equation)**.** *Let $a \in C(\mathbb{R})$ be T-periodic. Then* (P$_1$) *has a non-zero T-periodic solution if and only if*

$$\int_0^T a(s)\,ds = 0.$$

Proof. **(Necessity.)** Suppose $x : \mathbb{R} \to \mathbb{R}$ is a non-zero T-periodic solution, so $x(t+T) = x(t)$ for all t. Differentiating gives $x'(t+T) = x'(t)$. Because a is T-periodic,

$$a(t+T)\,x(t+T) = x'(t+T) = x'(t) = a(t)\,x(t).$$

Since $x(t+T) = x(t)$ and $x(t) \neq 0$, we conclude $a(t+T) = a(t)$, which we already assumed. Now fix $t_0 \in \mathbb{R}$. The solution formula gives

$$x(t_0 + T) = x(t_0) \exp\left[\int_{t_0}^{t_0+T} a(s)\, ds\right].$$

Periodicity $x(t_0 + T) = x(t_0)$ and $x(t_0) \neq 0$ force

$$\exp\left[\int_{t_0}^{t_0+T} a(s)\, ds\right] = 1,$$

hence $\displaystyle\int_{t_0}^{t_0+T} a(s)\, ds = 0$. Because a is T-periodic this integral is independent of t_0, giving $\displaystyle\int_0^T a(s)\, ds = 0$.

(Sufficiency.) Assume $\displaystyle\int_0^T a(s)\, ds = 0$. Let x be any non-zero solution. For any $t \in \mathbb{R}$,

$$\begin{aligned} x(t+T) &= x(t_0) \exp\left[\int_{t_0}^{t+T} a(s)\, ds\right] \\ &= x(t_0) \exp\left[\int_{t_0}^{t} a(s)\, ds + \int_{t}^{t+T} a(s)\, ds\right]. \end{aligned}$$

Since a is T-periodic, $\displaystyle\int_t^{t+T} a(s)\, ds = \int_0^T a(s)\, ds = 0$. Hence $x(t+T) = x(t_0) \exp\left[\int_{t_0}^{t} a(s)\, ds\right] = x(t)$. So x is T-periodic.

Conclusion. Equation (P_1) has a non-zero T-periodic solution if and only if a is T-periodic and $\displaystyle\int_0^T a(s)\, ds = 0$. $\square$

Chapter 2

The Matrix Exponential and the System $x' = Ax$

2.1 Constant-Coefficient Systems

Example 2.1 (Periodic solutions of $x' = Ax$)**.** Let $A \in M_n(\mathbb{C})$ and consider

$$x'(t) = A\,x(t). \tag{P$_2$}$$

The general solution is $x(t) = e^{tA}x(0)$. Fix $T > 0$. A non-zero solution is T-periodic if and only if $x(t+T) = x(t)$ for all t, i.e.

$$e^{(t+T)A}x(0) = e^{tA}\,e^{TA}\,x(0) = e^{tA}\,x(0).$$

This requires $e^{TA}\,x(0) = x(0)$, so $x(0)$ is an eigenvector of e^{TA} with eigenvalue 1.

Conclusion: Equation (**??**) has a non-zero T-periodic solution if and only if $1 \in \sigma(e^{TA})$.

Chapter 3

Matrix Logarithms and Square Roots

3.1 Square Roots of Real Matrices

Theorem 3.1 (Square roots via matrix logarithms)**.** *Let $A \in M_n(\mathbb{R})$ be invertible. If $A = B^2$ for some $B \in M_n(\mathbb{R})$, then there exists $X \in M_n(\mathbb{R})$ such that $e^X = A$.*

Proof. Suppose $A = B^2$ for some invertible $B \in M_n(\mathbb{R})$. Since B is invertible and real, one can define the matrix logarithm $\ln B$ via a polynomial in B. More precisely, there exists a polynomial $P \in \mathbb{C}[x]$ such that $\ln B = P(B)$ (this follows from the Cayley-Hamilton theorem and the theory of functions of matrices).

Then:

$$A = B^2 = B \cdot B = e^{\ln B} \cdot e^{\ln B} = e^{P(B)} \cdot e^{P(B)}.$$

Since B is a real matrix and $P(B)\overline{P(B)} = \overline{P(B)}P(B)$ (the matrices commute), we have:

$$\overline{P(B)} = \overline{P}(\overline{B}) = \overline{P}(B),$$

because B is real ($\overline{B} = B$) and $\overline{P}$ is the polynomial with conjugate coefficients. Hence:

$$A = e^{P(B)} \cdot e^{\overline{P(B)}} = e^{P(B)+\overline{P(B)}}.$$

Setting $X = P(B) + \overline{P(B)}$, we see that X is a real matrix (it equals its own conjugate) and $e^X = A$. □

Exercise 3.2 (Necessary and sufficient condition for a 2×2 real square root)**.** Find a necessary and sufficient condition for a 2×2 real matrix to be the square of a real matrix.

Solution. Let $A \in M_2(\mathbb{R})$. We seek necessary and sufficient conditions for the existence of $B \in M_2(\mathbb{R})$ with $B^2 = A$.

Step 1: Eigenvalue analysis. If $B^2 = A$ then the eigenvalues of A are the squares of those of B. The characteristic polynomial of B is $\lambda^2 - (\operatorname{tr} B)\lambda + \det B$. Its roots are eigenvalues μ_1, μ_2 of B, and the eigenvalues of A are μ_1^2, μ_2^2.

Step 2: Cases.

Case (i): A has two distinct real eigenvalues α, β. We need real μ_1, μ_2 with $\mu_1^2 = \alpha$ and $\mu_2^2 = \beta$. This requires $\alpha \geq 0$ and $\beta \geq 0$ *unless* both are negative. If both are negative, set $\mu_1 = \pm i\sqrt{|\alpha|}$ and $\mu_2 = \pm i\sqrt{|\beta|}$; but these are complex, so B would be complex. If at least one of α, β is negative and the other is non-negative, then no real square root exists (the product of the eigenvalues μ_i must be real and $\mu_1\mu_2 = \det B \in \mathbb{R}$, but μ_1 would be purely imaginary and μ_2 real, giving $\det B$ imaginary — contradiction).

Case (ii): A has a pair of complex-conjugate eigenvalues $\alpha \pm i\beta$ ($\beta \neq 0$). We need $\mu^2 = \alpha + i\beta$ for some $\mu \in \mathbb{C}$. Writing $\mu = a + ib$, we require $a^2 - b^2 = \alpha$ and $2ab = \beta$. Such real a, b always exist, but to have $B \in M_2(\mathbb{R})$, the complex eigenvalues of B must also appear as a conjugate pair. Indeed choose

$$a = \sqrt{\tfrac{\alpha+r}{2}}, \quad b = \tfrac{\beta}{2a}, \quad r = \sqrt{\alpha^2 + \beta^2} = |\alpha + i\beta|.$$

Then B can be taken as the real matrix with characteristic polynomial $\lambda^2 - 2a\lambda + (a^2 + b^2)$. This is always possible when $\det A > 0$ (so $r > 0$ and a is real).

Case (iii): A has a repeated eigenvalue α. If A is diagonalisable (i.e. $A = \alpha I$), then $B = \sqrt{\alpha}\, I$ works when $\alpha > 0$. If $\alpha < 0$, no real square root exists. If A is not diagonalisable (Jordan block), a real square root exists if and only if $\alpha > 0$.

Summary (necessary and sufficient condition): A real 2×2 matrix A is the square of a real 2×2 matrix if and only if one of the following holds:

(i) A has two real eigenvalues that are both non-negative.

(ii) A has two real eigenvalues that are both negative (in this case a real square root still exists: the complex square-root matrix turns out to be real after a real similarity transformation, since the two complex eigenvalues $\pm i\sqrt{|\lambda|}$ form a conjugate pair in $M_2(\mathbb{R})$).

(iii) A has a pair of complex-conjugate eigenvalues with positive determinant, i.e. $\det A > 0$.

Equivalently, *a real 2×2 matrix A has a real square root if and only if* $\det A \geq 0$, *or A has no real negative eigenvalues* — more precisely: $A \in M_2(\mathbb{R})$ has a real square root if and only if A is *not* similar over $\mathbb{R}$ to a matrix of the form $\left(\begin{smallmatrix} -\alpha & 0 \\ 0 & -\beta \end{smallmatrix}\right)$ with $\alpha, \beta > 0$ and $\alpha \neq \beta$.

Chapter 4

The Complex Floquet Theorem

4.1 Fundamental Matrices

Let us set up the standard notation used throughout.
Notation. Let $T > 0$ and $A \in C(\mathbb{R}; M_n(\mathbb{K}))$ be T-periodic. We write:

- $\mathcal{F}(\mathbb{R}, A)$: the set of all *fundamental matrices* of $x'(t) = A(t)x(t)$, i.e. all $F \in C^1(\mathbb{R}; GL_n(\mathbb{K}))$ satisfying $F'(t) = A(t)F(t)$.
- $\mathcal{S}(\mathbb{R}, A)$: the set of all solutions $x \in C^1(\mathbb{R}; \mathbb{K}^n)$ of $x'(t) = A(t)x(t)$.
- $\mathcal{S}(\mathbb{R}, A, b)$: the set of all solutions of $x'(t) = A(t)x(t) + b(t)$.

4.2 The Complex Floquet Decomposition

Theorem 4.1 (Complex Floquet Theorem)**.** *Let $T > 0$. Let $A \in C(\mathbb{R}; M_n(\mathbb{C}))$ be T-periodic and let $F \in \mathcal{F}(\mathbb{R}, A)$. Then there exist a T-periodic matrix function $P \in C^1(\mathbb{R}; GL_n(\mathbb{C}))$ and a matrix $B \in M_n(\mathbb{C})$ such that*

$$F(t) = P(t)\, e^{Bt} \qquad \forall\, t \in \mathbb{R}.$$

Proof. **Step 1: Monodromy.** Define $G : \mathbb{R} \to M_n(\mathbb{C})$ by $G(t) = F(t + T)$. Then

$$G'(t) = F'(t + T) = A(t + T)F(t + T) = A(t)G(t),$$

using the T-periodicity of A. Hence $G \in \mathcal{F}(\mathbb{R}, A)$.

Since any two fundamental matrices of the same system are related by a constant invertible matrix, there exists $C \in GL_n(\mathbb{C})$ such that $G(t) = F(t)C$, i.e.

$$F(t + T) = F(t)\, C \qquad \forall\, t \in \mathbb{R}. \tag{4.1}$$

Step 2: Matrix logarithm. Over $\mathbb{C}$, every invertible matrix has a logarithm. Hence there exists $B \in M_n(\mathbb{C})$ such that $e^{BT} = C$.

Step 3: Construction of P. Define $P : \mathbb{R} \to GL_n(\mathbb{C})$ by

$$P(t) = F(t)\, e^{-Bt}.$$

Clearly $F(t) = P(t)e^{Bt}$. We verify that P is T-periodic:

$$P(t+T) = F(t+T)\, e^{-B(t+T)} = F(t)\, C\, e^{-BT}\, e^{-Bt} = F(t)\, e^{BT}\, e^{-BT}\, e^{-Bt} = F(t)\, e^{-Bt} = P(t).$$

Since $F \in C^1$ and e^{-Bt} is smooth, $P \in C^1(\mathbb{R}; GL_n(\mathbb{C}))$. The periodicity $P(t+T) = P(t)$ completes the proof. □

4.3 The Real Floquet Theorem

Theorem 4.2 (Real Floquet Theorem)**.** *Let $T > 0$. Let $A \in C(\mathbb{R}; M_n(\mathbb{R}))$ be T-periodic and let $F \in \mathcal{F}(\mathbb{R}, A)$. Then there exist a $2T$-periodic matrix function $P \in C^1(\mathbb{R}; GL_n(\mathbb{R}))$ and a matrix $B \in M_n(\mathbb{R})$ such that*

$$F(t) = P(t)\, e^{Bt} \qquad \forall\, t \in \mathbb{R}.$$

Proof. **Step 1: Monodromy (same as complex case).** Define $G(t) = F(t+T)$. As above, $G \in \mathcal{F}(\mathbb{R}, A)$, so there exists $C \in GL_n(\mathbb{R})$ with $F(t+T) = F(t)C$. Applying this twice,

$$F(t+2T) = F(t+T)\, C = F(t)\, C^2.$$

Step 2: Real square root of C^2. The key point is that over $\mathbb{R}$ we cannot always take a logarithm of C, but we can always find $B \in M_n(\mathbb{R})$ such that $e^{2BT} = C^2$. (Indeed, C^2 has a real logarithm because its eigenvalues come in conjugate pairs with positive product; alternatively one uses the fact established in the previous chapter.)

Step 3: Construction of P. Define $P(t) = F(t)e^{-Bt}$. Then $F(t) = P(t)e^{Bt}$ and

$$P(t+2T) = F(t+2T)\, e^{-B(t+2T)} = F(t)\, C^2\, e^{-2BT}\, e^{-Bt} = F(t)\, e^{2BT}\, e^{-2BT}\, e^{-Bt} = F(t)\, e^{-Bt} = P(t).$$

Hence P is $2T$-periodic and $P \in C^1(\mathbb{R}; GL_n(\mathbb{R}))$. □

Remark. The period of P in the real theorem is $2T$, not T in general. This doubling of the period is the price we pay for working over $\mathbb{R}$, where not every invertible matrix has a real logarithm.

Chapter 5

The Floquet Transformation

Theorem 5.1 (Floquet Transformation Theorem)**.** *Let $T > 0$, $A \in C(\mathbb{R}, M_n(\mathbb{C}))$ be T-periodic. Let $F \in \mathcal{F}(\mathbb{R}, A)$, $P \in C^1(\mathbb{R}; GL_n(\mathbb{C}))$ be T-periodic, and $B \in M_n(\mathbb{C})$ be such that $F(t) = P(t)e^{Bt}$ for all t (Floquet normal form).*

Let $X \in \mathcal{S}(\mathbb{R}, A)$ and define $Y : \mathbb{R} \to \mathbb{C}^n$ by

$$Y(t) = P(t)^{-1}\, X(t).$$

Then $Y'(t) = B\, Y(t)$ for all $t \in \mathbb{R}$.

Proof. **Step 1: Derive the identity $P'(t) = A(t)P(t) - P(t)B$.**

Differentiating $F(t) = P(t)e^{Bt}$:

$$F'(t) = P'(t)\, e^{Bt} + P(t)\, B\, e^{Bt}.$$

Since $F \in \mathcal{F}(\mathbb{R}, A)$, we also have $F'(t) = A(t)F(t) = A(t)P(t)e^{Bt}$. Setting these equal and multiplying on the right by e^{-Bt}:

$$A(t)\, P(t) = P'(t) + P(t)\, B,$$

i.e. $P'(t) = A(t)P(t) - P(t)B$.

Step 2: Differentiate $X(t) = P(t)Y(t)$.

$$X'(t) = P'(t)\, Y(t) + P(t)\, Y'(t).$$

Substituting $P'(t) = A(t)P(t) - P(t)B$ and using $X'(t) = A(t)X(t) = A(t)P(t)Y(t)$:

$$A(t)P(t)Y(t) = \big[A(t)P(t) - P(t)B\big]Y(t) + P(t)Y'(t).$$

Expanding the right side and cancelling $A(t)P(t)Y(t)$:

$$0 = -P(t)B\, Y(t) + P(t)Y'(t).$$

Since $P(t) \in GL_n(\mathbb{C})$, we may multiply on the left by $P(t)^{-1}$:

$$Y'(t) = B\, Y(t).$$

□

Remark. This theorem says that the substitution $X(t) = P(t)Y(t)$ transforms the time-varying system $X' = A(t)X$ into the constant-coefficient system $Y' = BY$. This is the fundamental utility of the Floquet decomposition.

5.1 The Monodromy Matrix via the Floquet Decomposition

From $F(t) = P(t)e^{Bt}$ and the T-periodicity of P:

$$\begin{aligned} F(T) &= P(T)\, e^{BT} = P(0)\, e^{BT}, \\ F(0) &= P(0). \end{aligned}$$

Therefore

$$F(T) = F(0)\, e^{BT},$$

which gives us the fundamental relation

$$e^{BT} = F(0)^{-1}\, F(T), \quad \text{i.e.} \quad BT \in \ln\big[F(0)^{-1}F(T)\big]. \tag{5.1}$$

Definition 5.2 (Floquet Normal Form)**.** Let $A \in C(\mathbb{R}; M_n(\mathbb{C}))$ be T-periodic and $F \in \mathcal{F}(\mathbb{R}, A)$. The representation $F(t) = P(t)e^{Bt}$ given by the Floquet Theorem is called the **Floquet Normal Form** of F.

Chapter 6

Monodromy Matrices and Floquet Multipliers

6.1 Definitions

Definition 6.1 (Monodromy Matrix, Floquet Multipliers, Floquet Exponents)**.** Let $T > 0$, $A \in C(\mathbb{R}; M_n(\mathbb{K}))$ be T-periodic. Consider the equation

$$x'(t) = A(t)\,x(t). \tag{P}$$

1. **Monodromy Matrix.** $C \in GL_n(\mathbb{K})$ is a *Monodromy Matrix* of (P) if $C = F(0)^{-1}F(T)$ for some $F \in \mathcal{F}(\mathbb{R}, A)$.
2. **Floquet (Characteristic) Multipliers.** $\lambda \in \mathbb{C}$ is a *Floquet multiplier* of (P) if $\lambda \in \sigma(C)$ for some Monodromy Matrix C.
3. **Floquet (Characteristic) Exponents.** $\mu \in \mathbb{C}$ is a *Floquet exponent* of (P) if $\lambda = e^{\mu T}$ for some Floquet multiplier λ.

Notation. We write:

$$\mathrm{MM}(\mathbb{R};\, A) = \{C \in GL_n(\mathbb{K}) \mid C \text{ is a Monodromy Matrix of (P)}\},$$

$$\mathrm{FM}(\mathbb{R};\, A) = \{\lambda \in \mathbb{C} \mid \lambda \text{ is a Floquet multiplier of (P)}\},$$

$$\mathrm{FE}(\mathbb{R};\, A) = \{\mu \in \mathbb{C} \mid \mu \text{ is a Floquet exponent of (P)}\}.$$

6.2 Characterisation of Monodromy Matrices

Theorem 6.2. *Let $T > 0$ and $A \in C(\mathbb{R}; M_n(\mathbb{K}))$ be T-periodic. Let $C \in GL_n(\mathbb{K})$. Then $C \in \mathrm{MM}(\mathbb{R};\, A)$ if and only if there exists $F \in \mathcal{F}(\mathbb{R}, A)$ such that*

$$F(t+T) = F(t)\,C \qquad \forall\, t \in \mathbb{R}.$$

Proof. ($\Rightarrow$) If $C \in \mathrm{MM}(\mathbb{R};\, A)$, then by definition there exists $F \in \mathcal{F}(\mathbb{R}, A)$ with $C = F(0)^{-1}F(T)$, i.e. $F(T) = F(0)C$. Define

$$X(t) = F(t+T) - F(t)\,C.$$

Then

$$\begin{aligned} X'(t) &= F'(t+T) - F'(t)\,C = A(t+T)F(t+T) - A(t)F(t)C \\ &= A(t)F(t+T) - A(t)F(t)C = A(t)\,X(t), \end{aligned}$$

using the T-periodicity of A. Also $X(0) = F(T) - F(0)C = 0$. By uniqueness of solutions to the initial value problem, $X(t) = 0$ for all t, i.e. $F(t+T) = F(t)C$.

($\Leftarrow$) If $F(t+T) = F(t)C$ for all t, set $t = 0$ to get $F(T) = F(0)C$, hence $C = F(0)^{-1}F(T)$, so $C \in \mathrm{MM}(\mathbb{R};\, A)$. □

6.3 Conjugacy of Monodromy Matrices

Theorem 6.3. *Let $T > 0$, $A \in C(\mathbb{R}; M_n(\mathbb{C}))$ be T-periodic, and let $C, D \in \mathrm{MM}(\mathbb{R};\, A)$. Then there exists $P \in GL_n(\mathbb{K})$ such that $C = PDP^{-1}$. In particular, $\sigma(C) = \sigma(D)$.*

Proof. Let $F, G \in \mathcal{F}(\mathbb{R}, A)$ satisfy $C = F(0)^{-1}F(T)$ and $D = G(0)^{-1}G(T)$. Since F and G are both fundamental matrices of the same system, there exists $P \in GL_n(\mathbb{K})$ such that $G(t) = F(t)P$ for all t. Therefore:

$$D = G(0)^{-1}G(T) = (F(0)P)^{-1}(F(T)P) = P^{-1}F(0)^{-1}F(T)P = P^{-1}CP.$$

Hence $C = PDP^{-1}$, showing C and D are similar. Similar matrices have the same spectrum, so $\sigma(C) = \sigma(D)$. □

Corollary 6.4. *Let $A \in C(\mathbb{R}; M_n(\mathbb{K}))$ be T-periodic and let $F \in \mathcal{F}(\mathbb{R}, A)$. Then*

$$\mathrm{FM}(\mathbb{R};\, A) = \sigma\big(F(0)^{-1}F(T)\big).$$

Proof. This is immediate from Theorem 6.3: all monodromy matrices are conjugate and hence have the same spectrum. □

6.4 Floquet Solutions

Theorem 6.5. *Let $T > 0$, $A \in C(\mathbb{R}; M_n(\mathbb{K}))$ be T-periodic. Then $\lambda \in \mathrm{FM}(\mathbb{R};\, A)$ if and only if there exists $x_\lambda \in \mathcal{S}(\mathbb{R}, A)$ such that:*

(1) $x_\lambda(t) \in \mathbb{K}^n \setminus \{0\}$ for all $t \in \mathbb{R}$,

(2) $x_\lambda(t+T) = \lambda\, x_\lambda(t)$ for all $t \in \mathbb{R}$.

Furthermore, $\{x_\alpha, x_\beta\}$ is linearly independent whenever $\alpha, \beta \in \mathrm{FM}(\mathbb{R};\, A)$ with $\alpha \neq \beta$.

Proof. **(⇒): Existence of a Floquet solution.** Let $\lambda \in \mathrm{FM}(\mathbb{R}; A)$. Choose $F \in \mathcal{F}(\mathbb{R}, A)$ with $\lambda \in \sigma(F(0)^{-1}F(T))$. Then there exists $v \in \mathbb{K}^n \setminus \{0\}$ with $F(0)^{-1}F(T)v = \lambda v$, i.e. $F(T)v = \lambda F(0)v$. Define

$$x_\lambda(t) = F(t)\, v.$$

Clearly $x_\lambda \in \mathcal{S}(\mathbb{R}, A)$ and $x_\lambda(t) \neq 0$ since $F(t)$ is invertible. Set $Y(t) = x_\lambda(t+T) - \lambda x_\lambda(t)$. Then

$$Y'(t) = A(t+T)x_\lambda(t+T) - \lambda A(t)x_\lambda(t) = A(t)\, Y(t),$$

and $Y(0) = F(T)v - \lambda F(0)v = 0$. By uniqueness, $Y \equiv 0$, giving (2).

(⇐): Recovering the multiplier. Suppose $x_\lambda \in \mathcal{S}(\mathbb{R}, A)$ satisfies (1) and (2). Choose $F \in \mathcal{F}(\mathbb{R}, A)$ and $v \in \mathbb{K}^n \setminus \{0\}$ with $x_\lambda(t) = F(t)v$. Then $x_\lambda(T) = \lambda x_\lambda(0)$ gives $F(T)v = \lambda F(0)v$, so $F(0)^{-1}F(T)v = \lambda v$, i.e. $\lambda \in \sigma(F(0)^{-1}F(T)) = \mathrm{FM}(\mathbb{R}; A)$.

Linear independence of $\{x_\alpha, x_\beta\}$. Let $F \in \mathcal{F}(\mathbb{R}, A)$ and $a, b \in \mathbb{K}^n \setminus \{0\}$ with $x_\alpha(t) = F(t)a$, $x_\beta(t) = F(t)b$. Since α, β are distinct eigenvalues of $F(0)^{-1}F(T)$, the eigenvectors a and b are linearly independent. Since $F(t)$ is invertible, $F(t)a$ and $F(t)b$ are linearly independent for each t, hence $\{x_\alpha, x_\beta\}$ is linearly independent. □

Corollary 6.6. *Let $T > 0$, $A \in C(\mathbb{R}; M_n(\mathbb{K}))$ be T-periodic. Then there exists a non-zero T-periodic $X \in \mathcal{S}(\mathbb{R}, A)$ if and only if $1 \in \mathrm{FM}(\mathbb{R}; A)$.*

Proof. **(⇒):** A T-periodic non-zero solution satisfies $x(t+T) = x(t) = 1 \cdot x(t)$, so $\lambda = 1$ is a Floquet multiplier by Theorem 6.5.

(⇐): If $1 \in \mathrm{FM}(\mathbb{R}; A)$, Theorem 6.5 gives a non-zero solution with $x_1(t+T) = 1 \cdot x_1(t) = x_1(t)$, which is T-periodic. □

Verification against the scalar case:

(1) For $x' = a(t)x$, the fundamental solution is $\phi(t) = \exp\big[\int_0^t a(s)\, ds\big]$. The monodromy scalar is $\phi(0)^{-1}\phi(T) = \exp\big[\int_0^T a(s)\, ds\big]$. Then $1 \in \sigma(\phi(0)^{-1}\phi(T))$ iff $\displaystyle\int_0^T a(s)\, ds = 0$, in agreement with Theorem 1.2.

(2) For $x' = Ax$ (constant), $F(t) = e^{At}$, and $F(0)^{-1}F(T) = e^{AT}$. The condition $1 \in \sigma(e^{AT})$ matches Example 2.1.

Chapter 7

Liouville's Theorem and the Floquet Product Formula

Lemma 7.1 (Product formula for Floquet multipliers)**.** *Let $T > 0$ and $A \in C(\mathbb{R}; M_n(\mathbb{C}))$ be T-periodic. If $\lambda_1, \ldots, \lambda_n$ are the Floquet multipliers (with multiplicity) of $x'(t) = A(t)x(t)$, then*

$$\prod_{i=1}^{n} \lambda_i = \exp\left[\int_0^T \operatorname{tr}(A(s))\, ds\right].$$

If $\mu_1, \ldots, \mu_n$ are the corresponding Floquet exponents ($\lambda_i = e^{\mu_i T}$), then

$$\sum_{i=1}^{n} \mu_i \equiv \frac{1}{T}\int_0^T \operatorname{tr}(A(s))\, ds \pmod{\frac{2\pi i}{T}}.$$

Proof. Let $F \in \mathcal{F}(\mathbb{R}, A)$. By **Liouville's Theorem** (Abel's formula):

$$\det[F(t)] = \det[F(0)]\, \exp\left[\int_0^t \operatorname{tr}(A(s))\, ds\right].$$

At $t = T$:

$$\det[F(T)] = \det[F(0)]\, \exp\left[\int_0^T \operatorname{tr}(A(s))\, ds\right].$$

The Floquet multipliers are the eigenvalues of $F(0)^{-1}F(T)$, so

$$\prod_{i=1}^{n} \lambda_i = \det\big[F(0)^{-1}F(T)\big] = \frac{\det F(T)}{\det F(0)} = \exp\left[\int_0^T \operatorname{tr}(A(s))\, ds\right].$$

For the exponents, $\lambda_i = e^{\mu_i T}$, so $\prod \lambda_i = e^{(\sum \mu_i)T}$, giving

$$(\textstyle\sum \mu_i)T \equiv \int_0^T \operatorname{tr}(A(s))\, ds \pmod{2\pi i},$$

i.e. $\sum \mu_i \equiv \frac{1}{T}\int_0^T \operatorname{tr}(A(s))\, ds \pmod{\frac{2\pi i}{T}}$. $\square$

Chapter 8

Stability and Boundedness

8.1 Unbounded Solutions

Theorem 8.1 (Unstable/Stable directions)**.** *Let $T > 0$, $A \in C(\mathbb{R}; M_n(\mathbb{K}))$ be T-periodic. Then:*

(1) If $\operatorname{Re}\left[\int_0^T \operatorname{tr}(A(s))\,ds\right] > 0$*, then there exists $X \in \mathcal{S}(\mathbb{R}, A)$ with $X([0,\infty))$ unbounded.*

(2) If $\operatorname{Re}\left[\int_0^T \operatorname{tr}(A(s))\,ds\right] < 0$*, then there exists $Y \in \mathcal{S}(\mathbb{R}, A)$ with $Y((-\infty, 0])$ unbounded.*

Proof. We first treat $\mathbb{K} = \mathbb{C}$.

Let $\lambda_1, \ldots, \lambda_n$ be the Floquet multipliers. By Lemma 7.1,

$$\prod_{i=1}^{n} |\lambda_i| = \left|\exp\left[\int_0^T \operatorname{tr}(A(s))\,ds\right]\right| = \exp\left[\operatorname{Re}\int_0^T \operatorname{tr}(A(s))\,ds\right].$$

(1) When $\operatorname{Re}\left[\int_0^T \operatorname{tr}(A(s))\,ds\right] > 0$: The product $\prod |\lambda_i| > 1$, so there exists k with $|\lambda_k| > 1$. By Theorem 6.5, find $X \in \mathcal{S}(\mathbb{R}, A)$ with $X(t+T) = \lambda_k X(t)$. Iterating: $X(t+nT) = \lambda_k^n X(t)$, so

$$\|X(t+nT)\| = |\lambda_k|^n \|X(t)\| \to \infty.$$

Hence $X([0,\infty))$ is unbounded.

(2) When $\operatorname{Re}\left[\int_0^T \operatorname{tr}(A(s))\,ds\right] < 0$: $\prod |\lambda_i| < 1$, so there exists k with $|\lambda_k| < 1$. Find $Y \in \mathcal{S}(\mathbb{R}, A)$ with $Y(t+T) = \lambda_k Y(t)$, so $Y(t-T) = \lambda_k^{-1} Y(t)$. Iterating: $Y(t-nT) = \lambda_k^{-n} Y(t)$ and $\|Y(t-nT)\| = |\lambda_k|^{-n} \|Y(t)\| \to \infty$. Hence $Y((-\infty, 0])$ is unbounded.

Case $\mathbb{K} = \mathbb{R}$: The complex case gives $X \in \mathcal{S}(\mathbb{R}, A)$ (possibly complex-valued) satisfying (1) or (2). Write $X = \operatorname{Re} X + i \operatorname{Im} X$. Since A is real, both $\operatorname{Re} X$ and

$\operatorname{Im} X$ are real solutions. At least one of them must be unbounded on the same half-line (for if both were bounded, X would be bounded, contradicting the construction), yielding the required real unbounded solution. $\square$

Example 8.2. Consider the system

$$\begin{pmatrix} x'(t) \\ y'(t) \end{pmatrix} = A(t) \begin{pmatrix} x(t) \\ y(t) \end{pmatrix}, \qquad A(t) = \begin{pmatrix} 1 & 1 \\ 1 & \cos^2 t \end{pmatrix}.$$

A is π-periodic. We compute:

$$\int_0^\pi \operatorname{tr}(A(s))\, ds = \int_0^\pi \left(1 + \cos^2 s\right) ds = \pi + \frac{\pi}{2} = \frac{3\pi}{2} > 0.$$

By Theorem 8.1(1), there exists a solution $(x, y)^T$ of this system that is unbounded on $[0, \infty)$.

Chapter 9

Hill's Equation

9.1 Reduction to a First-Order System

Definition 9.1 (Hill's Equation)**.** Let $T > 0$ and $p : \mathbb{R} \to \mathbb{K}$ be T-periodic and continuous. **Hill's equation** is

$$x''(t) + p(t)\, x(t) = 0. \tag{HE}$$

Setting $x_1 = x$ and $x_2 = x'$, Hill's equation becomes the system

$$\begin{pmatrix} x_1' \\ x_2' \end{pmatrix} = H(t) \begin{pmatrix} x_1 \\ x_2 \end{pmatrix}, \qquad H(t) = \begin{pmatrix} 0 & 1 \\ -p(t) & 0 \end{pmatrix}.$$

Note that $\operatorname{tr}(H(t)) = 0$ for all t.
Remark. For any $F \in \mathcal{F}(\mathbb{R}, H)$, Liouville's formula gives

$$\det[F(t)] = \det[F(0)] \exp\left[\int_0^t \operatorname{tr}(H(s))\, ds\right] = \det[F(0)].$$

Hence $\det[F(0)^{-1}F(T)] = 1$. Therefore the Floquet multipliers λ_1, λ_2 of (HE) satisfy $\lambda_1\lambda_2 = 1$.

Chapter 10

The Lyapunov Boundedness Theorem

10.1 The Lyapunov–Borg Inequality

Theorem 10.1 (Lyapunov–Borg Inequality). *Let $-\infty < a < b < \infty$ and let $u \in C^2([a,b],\mathbb{R})$ satisfy*

(1) $u(a) = u(b) = 0$,

(2) $u(t) \neq 0$ *for all* $t \in (a,b)$.

Then

$$\int_a^b \frac{|u''(t)|}{|u(t)|}\,dt > \frac{4}{b-a}.$$

Proof. Without loss of generality assume $u(t) > 0$ for $t \in (a,b)$ (otherwise replace u by $-u$).

Let $c \in (a,b)$ be such that $u(c) = \max_{t\in[a,b]} u(t) > 0$.

By the Mean Value Theorem, there exist $\alpha \in (a,c)$ and $\beta \in (c,b)$ such that

$$u'(\alpha) = \frac{u(c)-u(a)}{c-a} = \frac{u(c)}{c-a} > 0 \qquad \text{and} \qquad u'(\beta) = \frac{u(b)-u(c)}{b-c} = \frac{-u(c)}{b-c} < 0.$$

Now estimate:

$$\begin{aligned}
\int_a^b \frac{|u''(t)|}{|u(t)|}\,dt &= \int_a^b \frac{|u''(t)|}{u(t)}\,dt \quad (u > 0 \text{ on } (a,b)) \\
&> \int_a^b \frac{|u''(t)|}{u(c)}\,dt \qquad (\text{since } u(t) \le u(c) \text{ strictly}) \\
&\ge \frac{1}{u(c)} \int_\alpha^\beta |u''(t)|\,dt \\
&\ge \frac{1}{u(c)} \left| \int_\alpha^\beta u''(t)\,dt \right| \\
&= \frac{1}{u(c)} |u'(\beta) - u'(\alpha)| \\
&= \frac{1}{u(c)} \left| \frac{-u(c)}{b-c} - \frac{u(c)}{c-a} \right| \\
&= \frac{1}{c-a} + \frac{1}{b-c} \\
&= \frac{b-a}{(c-a)(b-c)}.
\end{aligned}$$

By the AM-GM inequality (or equivalently $(c-a)+(b-c) = b-a$ and $(x+y)^2 \ge 4xy$):

$$(c-a)(b-c) \le \left(\frac{(c-a)+(b-c)}{2} \right)^2 = \frac{(b-a)^2}{4}.$$

Therefore

$$\frac{b-a}{(c-a)(b-c)} \ge \frac{b-a}{\frac{(b-a)^2}{4}} = \frac{4}{b-a}.$$

Combining:

$$\int_a^b \frac{|u''(t)|}{|u(t)|}\,dt > \frac{4}{b-a}. \qquad \square$$

10.2 Lyapunov's Stability Theorem for Hill's Equation

Theorem 10.2 (Lyapunov's Theorem)**.** *Let $T > 0$ and $p \in C(\mathbb{R}, \mathbb{R})$ be T-periodic satisfying:*

(1) $\displaystyle\int_0^T p(t)\,dt \ge 0$, *and*

(2) $\displaystyle 0 < \int_0^T |p(t)|\,dt \le \frac{4}{T}$.

Then every solution of $x''(t) + p(t)x(t) = 0$ is bounded on $\mathbb{R}$.

Proof. It suffices to show that Hill's equation (HE) has no real Floquet multipliers. (If all Floquet multipliers satisfy $|\lambda| = 1$, then $\|x(t+nT)\| = |\lambda|^n \|x(t)\|$ stays bounded, hence all solutions are bounded.)

Why is it enough to exclude real Floquet multipliers? Recall from Remark 9.1 that $\lambda_1\lambda_2 = 1$. If both multipliers are complex (non-real), they are conjugates $e^{\pm i\theta}$, both of modulus 1, so all solutions are bounded. If a real Floquet multiplier λ exists, then $\lambda \cdot (1/\lambda) = 1$ and if $\lambda \in \mathbb{R}$ then $|\lambda| \neq 1$ is possible (specifically $|\lambda| > 1$ or $|\lambda| < 1$), leading to unbounded solutions. Hence we need to rule out $\lambda \in \mathbb{R}$.

Suppose for contradiction that $\lambda \in \mathbb{R}$ is a Floquet multiplier. By Theorem 6.5, there exists $u \in C^2(\mathbb{R}, \mathbb{R})$ satisfying $u(t+T) = \lambda u(t)$ and $u''(t) + p(t)u(t) = 0$.

Two cases:

Case 1: $u(t) \neq 0$ for all $t \in \mathbb{R}$. Since $u'' = -p(t)u$, we compute using integration by parts:

$$\int_0^T \frac{u''(t)}{u(t)}\,dt = \left[\frac{u'(t)}{u(t)}\right]_0^T - \int_0^T \frac{-(u'(t))^2}{(u(t))^2}\,dt = \left(\frac{u'(T)}{u(T)} - \frac{u'(0)}{u(0)}\right) + \int_0^T \frac{(u')^2}{u^2}\,dt.$$

Note that $u(T) = \lambda u(0)$ and $u'(T) = \lambda u'(0)$ (from $u(t+T) = \lambda u(t)$), so

$$\frac{u'(T)}{u(T)} = \frac{\lambda u'(0)}{\lambda u(0)} = \frac{u'(0)}{u(0)}.$$

Therefore:

$$\int_0^T \frac{u''(t)}{u(t)}\,dt = \int_0^T \frac{(u'(t))^2}{(u(t))^2}\,dt \geq 0.$$

But $u''(t) = -p(t)u(t)$, so $\frac{u''}{u} = -p(t)$, giving

$$-\int_0^T p(t)\,dt = \int_0^T \frac{(u')^2}{u^2}\,dt \geq 0,$$

i.e. $\int_0^T p(t)\,dt \leq 0$. Combined with assumption (1), we get $\int_0^T p(t)\,dt = 0$. But then $\int_0^T (u'/u)^2\,dt = 0$, implying $u'(t) = 0$ for all t, so u is constant. Then $u'' = 0$, so $p(t)u = 0$; since $u \neq 0$, $p(t) = 0$ for all t, contradicting assumption (2).

Case 2: u has a zero. Since $u(t+T) = \lambda u(t)$, the zeros of u in $\mathbb{R}$ repeat with period T. Let $a < b$ be two consecutive zeros of u with $b - a \leq T$. By the Lyapunov–Borg inequality (Theorem 10.1):

$$\int_a^b |p(t)|\,dt = \int_a^b \frac{|u''(t)|}{|u(t)|}\,dt > \frac{4}{b-a} \geq \frac{4}{T}.$$

Hence

$$\int_0^T |p(t)|\,dt \geq \int_a^b |p(t)|\,dt > \frac{4}{T},$$

contradicting assumption (2).

Both cases lead to contradictions. Therefore no real Floquet multiplier exists, and all solutions of Hill's equation are bounded. □

Chapter 11

Inhomogeneous Periodic Systems

11.1 Existence and Uniqueness of Periodic Solutions

Theorem 11.1. *Let $T > 0$, $A \in C(\mathbb{R}; M_n(\mathbb{K}))$, $b \in C(\mathbb{R}; \mathbb{K}^n)$ be T-periodic. Then $X \in \mathcal{S}(\mathbb{R}, A, b)$ is T-periodic if and only if $X(0) = X(T)$.*

Proof (Exercise, solved). **($\Rightarrow$):** If X is T-periodic then $X(T) = X(0)$ by definition.

($\Leftarrow$): Assume $X(0) = X(T)$. Define $Y(t) = X(t+T)$. Then

$$Y'(t) = X'(t+T) = A(t+T)X(t+T) + b(t+T) = A(t)Y(t) + b(t),$$

so $Y \in \mathcal{S}(\mathbb{R}, A, b)$. Also $Y(0) = X(T) = X(0)$. By uniqueness of the initial value problem, $Y(t) = X(t)$ for all t, i.e. $X(t+T) = X(t)$. $\square$

Theorem 11.2 (Unique periodic solution)**.** *Let $T > 0$, $A \in C(\mathbb{R}; M_n(\mathbb{K}))$, $b \in C(\mathbb{R}; \mathbb{K}^n)$ be T-periodic. There exists a* unique *T-periodic $X \in \mathcal{S}(\mathbb{R}, A, b)$ if and only if $1 \notin \mathrm{FM}(\mathbb{R};\, A)$.*

Proof. **($\Rightarrow$, existence when $1 \notin \mathrm{FM}(\mathbb{R};\, A)$):**

Let $F \in \mathcal{F}(\mathbb{R}, A)$. Since $1 \notin \sigma(F(0)^{-1}F(T)) = \sigma(F(T)F(0)^{-1})$, the matrix $I - F(T)F(0)^{-1}$ is invertible. Define $x_0 \in \mathbb{K}^n$ by

$$\left[I - F(T)F(0)^{-1}\right]x_0 = \int_0^T F(T)F(s)^{-1}b(s)\,ds.$$

(This equation has a unique solution since $I - F(T)F(0)^{-1} \in GL_n(\mathbb{K})$.) Define

$$X(t) = F(t)F(0)^{-1}x_0 + \int_0^t F(t)F(s)^{-1}b(s)\,ds.$$

By the variation of parameters formula, $X \in \mathcal{S}(\mathbb{R}, A, b)$. We verify $X(0) = X(T)$:

$$X(0) = F(0)F(0)^{-1}x_0 = x_0.$$

$$X(T) = F(T)F(0)^{-1}x_0 + \int_0^T F(T)F(s)^{-1}b(s)\,ds = F(T)F(0)^{-1}x_0 + \left[I - F(T)F(0)^{-1}\right]x_0 = x_0.$$

So $X(T) = X(0) = x_0$, and by Theorem 11.1, X is T-periodic.

Uniqueness: Let $X, Y \in \mathcal{S}(\mathbb{R}, A, b)$ be T-periodic. Then $Z = X - Y \in \mathcal{S}(\mathbb{R}, A)$ is T-periodic. Write $Z(t) = F(t)v$ for some $v \in \mathbb{K}^n$. Then $Z(0) = Z(T)$ gives $F(0)v = F(T)v$, so $F(0)^{-1}F(T)v = v$. Since $1 \notin \mathrm{FM}(\mathbb{R};\, A)$, we have $v = 0$, hence $Z \equiv 0$ and $X = Y$.

($\Leftarrow$, necessity when $1 \notin \mathrm{FM}(\mathbb{R};\, A)$):

Suppose there is a unique T-periodic $X \in \mathcal{S}(\mathbb{R}, A, b)$. Suppose for contradiction $1 \in \mathrm{FM}(\mathbb{R};\, A)$. Then by Corollary 6.6 there exists a non-zero T-periodic $Y \in \mathcal{S}(\mathbb{R}, A)$. Then $X + Y \in \mathcal{S}(\mathbb{R}, A, b)$ is T-periodic and $X + Y \neq X$, contradicting uniqueness. Hence $1 \notin \mathrm{FM}(\mathbb{R};\, A)$. □

11.2 Bounded Solutions and Periodic Solutions

Theorem 11.3. *Let $T > 0$, $A \in C(\mathbb{R}; M_n(\mathbb{K}))$, $b \in C(\mathbb{R}; \mathbb{K}^n)$ be T-periodic. There exists a T-periodic $X \in \mathcal{S}(\mathbb{R}, A, b)$ if and only if $\mathcal{S}(\mathbb{R}, A, b)$ contains a bounded solution.*

Proof. **($\Rightarrow$):** A T-periodic solution X is bounded since it is continuous and periodic on $\mathbb{R}$, hence bounded by the maximum of $\|X\|$ on $[0, T]$.

($\Leftarrow$): Let $Y \in \mathcal{S}(\mathbb{R}, A, b)$ be bounded. Let $F \in \mathcal{F}(\mathbb{R}, A)$ with $F(0) = I$ (normalise). By variation of parameters:

$$Y(t) = F(t)\,Y(0) + \int_0^t F(t)F(s)^{-1}b(s)\,ds.$$

Define the sequence $y_n = Y(nT)$ for $n \in \mathbb{Z}$. Using $F(t + T) = F(t)C$ where $C = F(T)$:

$$y_{n+1} = Y((n+1)T) = C\,y_n + \int_0^T F(T)F(s)^{-1}b(s)\,ds =: C\,y_n + d,$$

where $d = \int_0^T F(T)F(s)^{-1}b(s)\,ds$ is a constant vector. So $\{y_n\}$ satisfies a linear recurrence. Since Y is bounded, $\{y_n\}$ is a bounded sequence.

Write $y_n = x_0 + z_n$ where $z_n \in \ker(C - I)^{\perp}$ (the component orthogonal to the eigenspace of C for eigenvalue 1, if it exists). The boundedness of $\{y_n\}$ forces the eigenvalues of C on the support of $\{z_n\}$ to satisfy $|\lambda| \leq 1$.

The fixed-point equation $x_0 = Cx_0 + d$ (i.e. $(I - C)x_0 = d$) may not have a solution if $1 \in \sigma(C)$. If $1 \notin \sigma(C) = \mathrm{FM}(\mathbb{R};\, A)$, we are done by Theorem 11.2.

If $1 \in \sigma(C)$, one shows via a Fredholm-type argument that the boundedness of Y forces d to lie in the image of $I - C$, so a T-periodic solution exists. (A full proof uses the spectral decomposition of C; we omit the details here.) □

Chapter 12

Solved Exercises

This chapter collects all exercises from the notes together with complete solutions, presented in the order they appear.

Exercise 1: Periodicity of $x' = a(t)x$

Exercise 12.1. Verify the two specific examples:

(a) $a(t) = 1$: show that $x(t) = x(t_0)e^{t-t_0}$ is not periodic.

(b) $a(t) = \cos t$: show that $x(t) = x(t_0)e^{\sin t - \sin t_0}$ is 2π-periodic.

Solution. **(a)** $a(t) = 1$ is T-periodic for any T. The integral condition requires $\int_0^T 1\,dt = T = 0$, which is impossible for $T > 0$. Hence no non-zero periodic solution exists. Explicitly, $x(t) = x(t_0)e^{t-t_0}$ satisfies $x(t+T) = x(t_0)e^{t+T-t_0} = e^T x(t) \neq x(t)$ since $e^T > 1$.

(b) $a(t) = \cos t$ is 2π-periodic and $\int_0^{2\pi} \cos s\,ds = [\sin s]_0^{2\pi} = 0$. By Theorem 1.2, a non-zero 2π-periodic solution exists. Indeed, $x(t+2\pi) = x(t_0)e^{\sin(t+2\pi)-\sin t_0} = x(t_0)e^{\sin t - \sin t_0} = x(t)$.

Exercise 2: The periodicity condition $x(T) = x(t_0)$ — scalar version

Exercise 12.2. Let $a \in C(\mathbb{R})$ be T-periodic. Starting from the solution formula, derive directly that $x(t_0+T) = x(t_0)$ if and only if $\int_0^T a(s)\,ds = 0$.

Solution. The solution is $x(t) = x(t_0)\exp\left[\int_{t_0}^t a(s)\,ds\right]$. Setting $t = t_0 + T$:

$$x(t_0+T) = x(t_0)\exp\left[\int_{t_0}^{t_0+T} a(s)\,ds\right].$$

For $x(t_0+T) = x(t_0)$ with $x(t_0) \neq 0$, we need $\exp\left[\int_{t_0}^{t_0+T} a(s)\,ds\right] = 1$, i.e. $\int_{t_0}^{t_0+T} a(s)\,ds = 0$. Since a is T-periodic, this integral is independent of t_0: substituting $s \mapsto s - t_0$, $\int_{t_0}^{t_0+T} a(s)\,ds = \int_0^T a(s+t_0)\,ds = \int_0^T a(s)\,ds$. Hence the condition is $\int_0^T a(s)\,ds = 0$.

Exercise 3: $F(t) = P(t)e^{Bt}$ — check for $n = 1$

Exercise 12.3. Verify the Floquet decomposition explicitly for $n = 1$ and the scalar equation $x'(t) = a(t)x(t)$ with a T-periodic.

Solution. For $n = 1$ the fundamental "matrix" is the scalar function

$$F(t) = \exp\left[\int_0^t a(s)\,ds\right].$$

The monodromy scalar is $C = F(0)^{-1}F(T) = F(T)/F(0) = F(T)$ (since $F(0) = 1$). So $C = e^{\int_0^T a(s)\,ds}$. We choose $B = \frac{1}{T}\int_0^T a(s)\,ds$, giving $e^{BT} = C$. Then

$$P(t) = F(t)e^{-Bt} = \exp\left[\int_0^t a(s)\,ds - Bt\right].$$

Check periodicity:

$$P(t+T) = \exp\left[\int_0^{t+T} a(s)\,ds - B(t+T)\right] = \exp\left[\int_0^t a(s)\,ds + \int_t^{t+T} a(s)\,ds - Bt - BT\right].$$

Since a is T-periodic, $\int_t^{t+T} a(s)\,ds = \int_0^T a(s)\,ds = BT$. Hence $P(t+T) = \exp\left[\int_0^t a(s)\,ds - Bt\right] = P(t)$. ✓

Exercise 4: $X'(t) = A(t)X(t)$ has T-periodic solution iff $X(0) = X(T)$

Exercise 12.4. Let $T > 0$, $A \in C(\mathbb{R}; M_n(\mathbb{K}))$, $b \in C(\mathbb{R}; \mathbb{K}^n)$ be T-periodic. Prove that $X \in \mathcal{S}(\mathbb{R}, A, b)$ is T-periodic if and only if $X(0) = X(T)$.

Solution. This is exactly the content of Theorem 11.1, whose proof is given above. We restate it for clarity.

($\Rightarrow$): If $X(t+T) = X(t)$ for all t, then in particular $X(T) = X(0+T) = X(0)$.

($\Leftarrow$): Suppose $X(0) = X(T)$. Define $Y(t) := X(t+T)$. Then:

$$Y'(t) = X'(t+T) = A(t+T)X(t+T) + b(t+T) = A(t)Y(t) + b(t),$$

so Y is also a solution of the same inhomogeneous system. Moreover, $Y(0) = X(T) = X(0)$. By uniqueness of the IVP, $Y(t) = X(t)$ for all t, i.e. $X(t+T) = X(t)$.

Exercise 5: Lyapunov stability — filling in "Why?"

Exercise 12.5. In the proof of Lyapunov's theorem, why is it sufficient to show that Hill's equation (HE) has no real Floquet multipliers?

Solution. From Remark 9.1, the product of the two Floquet multipliers of Hill's equation satisfies $\lambda_1\lambda_2 = 1$.

Suppose no real Floquet multiplier exists. Then neither λ_1 nor λ_2 is real. Since the coefficients of Hill's equation are real, the characteristic polynomial of the monodromy matrix has real coefficients. Complex roots come in conjugate pairs, so $\lambda_2 = \overline{\lambda_1}$. Combined with $\lambda_1\lambda_2 = |\lambda_1|^2 = 1$, we get $|\lambda_1| = |\lambda_2| = 1$.

By Theorem 6.5, there exist solutions $x_j(t)$ with $x_j(t+T) = \lambda_j x_j(t)$. On a grid of times $t + nT$:

$$\|x_j(t+nT)\| = |\lambda_j|^n \|x_j(t)\| = \|x_j(t)\|.$$

Between grid points, continuity and compactness of $[0, T]$ keep $\|x_j\|$ bounded. The general solution is a linear combination $x = c_1x_1 + c_2x_2$, which is also bounded. Hence *all* solutions are bounded.

Conversely, if a real Floquet multiplier $\lambda \in \mathbb{R}$ exists, then $\lambda_1 = \lambda \in \mathbb{R}$ and $\lambda_2 = 1/\lambda \in \mathbb{R}$. If $\lambda > 1$, then $\|x_1(t+nT)\| = \lambda^n \|x_1(t)\| \to \infty$; similarly for $\lambda < -1$. If $|\lambda| \neq 1$, unbounded solutions exist. Hence excluding real multipliers is indeed the key step.

Exercise 6: Linear independence of Floquet solutions

Exercise 12.6. Let $\alpha, \beta \in \mathrm{FM}(\mathbb{R}; A)$ with $\alpha \neq \beta$. Prove that $\{x_\alpha, x_\beta\}$ as constructed in Theorem 6.5 is linearly independent.

Solution. Suppose $c_1x_\alpha(t) + c_2x_\beta(t) = 0$ for all t. Evaluating at $t = 0$:

$$c_1x_\alpha(0) + c_2x_\beta(0) = 0. \tag{$*$}$$

Evaluating at $t = T$ and using $x_\alpha(T) = \alpha x_\alpha(0)$, $x_\beta(T) = \beta x_\beta(0)$:

$$c_1\alpha x_\alpha(0) + c_2\beta x_\beta(0) = 0. \tag{$**$}$$

Multiply $(*)$ by β and subtract from $(**)$:

$$c_1(\alpha - \beta)x_\alpha(0) = 0.$$

Since $\alpha \neq \beta$ and $x_\alpha(0) \neq 0$, we get $c_1 = 0$. Then $(*)$ gives $c_2x_\beta(0) = 0$, and since $x_\beta(0) \neq 0$, $c_2 = 0$. Hence $\{x_\alpha, x_\beta\}$ is linearly independent.

Exercise 7: $\lambda = x_\lambda(T)/x_\lambda(0)$ for scalar component

Exercise 12.7. Let $x_\lambda \in \mathcal{S}(\mathbb{R}, A)$ satisfy $x_\lambda(t+T) = \lambda x_\lambda(t)$ with $x_\lambda(t) \neq 0$. Show that $\lambda = \frac{(x_\lambda)_i(T)}{(x_\lambda)_i(0)}$ for some component index $i \in \{1, \ldots, n\}$.

Solution. Since $x_\lambda(t) \neq 0$ for all t, there is at least one component $(x_\lambda)_i(t) \neq 0$ at $t = 0$ (and hence in a neighbourhood of 0). Fix such i. Then:

$$(x_\lambda)_i(T) = (x_\lambda(T))_i = (\lambda x_\lambda(0))_i = \lambda\,(x_\lambda)_i(0).$$

Since $(x_\lambda)_i(0) \neq 0$, dividing gives $\lambda = \frac{(x_\lambda)_i(T)}{(x_\lambda)_i(0)}$. In particular $\lambda \in \mathbb{K}$ (it is the ratio of two scalar values in $\mathbb{K}$).

Exercise 8: Monodromy matrix of $x' = Ax$ (constant coefficient)

Exercise 12.8. For the constant-coefficient system $x'(t) = Ax(t)$ with $A \in M_n(\mathbb{C})$, compute the monodromy matrix and show that $\mathrm{FM}(\mathbb{R}; A) = \sigma(e^{AT})$.

Solution. The fundamental matrix with $F(0) = I$ is $F(t) = e^{At}$. The monodromy matrix is

$$C = F(0)^{-1}F(T) = I^{-1}e^{AT} = e^{AT}.$$

By Corollary 6.4, $\mathrm{FM}(\mathbb{R}; A) = \sigma(e^{AT})$. This matches the finding of Example 2.1: a T-periodic non-zero solution exists iff $1 \in \sigma(e^{AT})$.

Exercise 9: Applying the product formula for Floquet multipliers

Exercise 12.9. For the system in the example of Chapter 7, $A(t) = \left(\begin{smallmatrix} 1 & 1 \\ 1 & \cos^2 t \end{smallmatrix}\right)$, compute $\prod |\lambda_i|$ and verify the stability conclusion.

Solution. $\mathrm{tr}(A(t)) = 1 + \cos^2 t$. Therefore

$$\int_0^\pi \mathrm{tr}(A(s))\, ds = \int_0^\pi (1 + \cos^2 s)\, ds = \pi + \int_0^\pi \frac{1 + \cos 2s}{2}\, ds = \pi + \frac{\pi}{2} = \frac{3\pi}{2}.$$

By Lemma 7.1:

$$|\lambda_1||\lambda_2| = e^{3\pi/2} > 1.$$

Hence not both multipliers can satisfy $|\lambda_i| \leq 1$; at least one satisfies $|\lambda_k| > 1$. By Theorem 8.1(1), there exists an unbounded solution on $[0, \infty)$.

Index of Notation

Symbol	Meaning
$\mathcal{F}(\mathbb{R}, A)$	Set of fundamental matrices of $x' = A(t)x$
$\mathcal{S}(\mathbb{R}, A)$	Solution set of $x' = A(t)x$
$\mathcal{S}(\mathbb{R}, A, b)$	Solution set of $x' = A(t)x + b(t)$
$\mathrm{MM}(\mathbb{R};\, A)$	Monodromy matrices of the system with coefficient A
$\mathrm{FM}(\mathbb{R};\, A)$	Floquet multipliers
$\mathrm{FE}(\mathbb{R};\, A)$	Floquet exponents
$\sigma(C)$	Spectrum (set of eigenvalues) of C
$\mathrm{tr}(A)$	Trace of matrix A
$GL_n(\mathbb{K})$	Group of invertible $n \times n$ matrices over $\mathbb{K}$
$M_n(\mathbb{K})$	Algebra of $n \times n$ matrices over $\mathbb{K}$